Guoyuan Lyu

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RESEARCH INTERESTS

My previous work has focused on measurement-driven error diagnosis and closed-loop compensation in ultraprecision optical manufacturing. I aim to generalize this framework to advanced manufacturing and intelligent inspection by integrating in-situ metrology, physical process modeling, adaptive control, and agent-enabled engineering automation.

EDUCATION

Sep. 2025 – Present M.Eng. in Tianjin University, Instrument Science and Technology, Tianjin, China. Expected Graduation: Jun. 2028. Recommended for graduate admission. Advisor: Prof. Xiaodong Zhang

Sep. 2021 – Jun. 2025 B.Eng. in Tianjin University, Measurement and Control Technology and Instruments, Tianjin, China. GPA: 3.73 / 4.0

RESEARCH AND PROJECT EXPERIENCE

High-Uniformity Ultra-Precision Diamond Turning of Single-Crystal Germanium Microlens Arrays

Master's Thesis project | Government-Funded Research Project, Sep. 2025 – Present

A closed-loop compensatory manufacturing process for monolithic multi-mirror systems guided by optical performance

Lab-Initiated Research Project, Dec. 2025 – Aug. 2026

- Led the experimental implementation and metrology of a closed-loop compensation process for a monolithic dual-reflector optical system manufactured by single-point diamond turning.
- Developed and executed a Zemax ZOS-API workflow to simulate the propagation of measured surface errors and evaluate candidate compensation strategies in terms of system-level optical performance.
- Processed and analyzed machining, surface-metrology, and optical-performance data to evaluate the effectiveness of the compensation method. Reduced the system wavefront error RMS by 90% after compensation iteration, experimentally validating the effectiveness of the optical-performance-guided compensation framework.
- Consolidated the simulation and experimental results and led the preparation of the resulting co-first-author paper.

Compensation Single-Point Diamond Turning of Large Aperture Biconic Optical Surfaces

Industry-Facing Engineering Project, Apr. 2026 – Jul. 2026

- Led the diagnosis of systematic surface-form errors in the production machining of biconic optical components and established corresponding compensation strategies.
- Developed feature-specific surface-error fitting algorithms for different error morphologies, enabling accurate representation of measured form errors with low fitting residuals.
- Generated compensated tool paths for biconic surfaces with varying design parameters and supported their implementation in ultra-precision diamond-turning production.
- Developed the core dynamic-link libraries for error-source identification, surface-error fitting, and compensated tool-path generation, which were integrated into a production-oriented software system delivered for routine use by laboratory engineers.

Development of a Coherence Scanning Interferometry Probe for Optical Profilometry

Industry-Sponsored Project, Nov. 2024 – Nov. 2025

- Developed dynamic-link libraries for instrument control and software integration, including image acquisition, scanning control, topography reconstruction, data reprocessing, and surface-roughness evaluation.
- Integrated the algorithmic and software components with the probe hardware and carried out hands-on testing, contributing to the successful delivery of a functional CSI probe for an optical profilometer.
- Participated in the component selection of a compact CSI probe integrating broadband illumination, piezoelectric vertical scanning, an interferometric objective, and industrial-camera acquisition, building on the laboratory's earlier hardware development.
- Independently designed and conducted exploratory microsphere-assisted CSI experiments for enhanced lateral-resolution measurement and identified limitations in microsphere manipulation, experimental stability, and forward-model validation.

PUBLICATIONS

- Lei Liu*, **Guoyuan Lyu***, Zexiao Li, Fengzhou Fang, Xiaodong Zhang (*co-first authors), A closed-loop compensatory manufacturing process for monolithic multi-mirror systems guided by optical performance, Journal of Manufacturing Processes, Volume 179, 373–388, 2026.
  DOI
- **Guoyuan Lyu**, Qian Liu, Zexiao Li, Weisheng Cheng, Junge Xu, Xiaodong Zhang, Coherence scanning interferometric measurement of advanced optics: A review, Optics & Laser Technology, Volume 204, Part A, 115956, 2026.
  DOI · Journal Page
- Weisheng Cheng, Shuangxiong Yin, Yongxu Xiang, **Guoyuan Lyu**, Jun Yang, Zexiao Li, Xiaodong Zhang, Key technologies of a FOUP-integrated spectral confocal module for wafer warpage measurement, Optics and Lasers in Engineering, Volume 201, 109733, 2026.
  DOI · Journal Page
- Weisheng Cheng, Zexiao Li, Juncheng Yu, Fuqi Sun, **Guoyuan Lyu** and Xiaodong Zhang, Bionic-inspired intelligent detection of surface defects on semiconductor silicon wafers, Measurement Science and Technology, Volume 37, 255204, 2026.
  DOI · Journal Page

TECHNICAL SKILLS

- Programming and development tools: MATLAB, C++, CMake, C#, Python, Git
- Simulation software: Zemax OpticStudio, ZOS-API, VirtualLab Fusion
- Experimental and engineering tools: Ultra-precision diamond turning lathe, Optical profilometer (LUPHOScan), Coherence scanning interferometer, Laser interferometer, Confocal microscope, Industrial cameras, Motion platforms, Piezoelectric actuators
- Technical writing: Academic manuscripts (review articles and original research papers), Patent documents, Project reports, Research proposals and Academic presentations

ENGLISH PROFICIENCY

IELTS Academic 7.5 (Listening 8.0 | Reading 8.0 | Writing 7.0 | Speaking 6.5)

Last updated: 2026-11-09